Supplementary material

The Newcastle-Ottawa Scale (NOS) for Assessing the Quality of Studies Included into Present Meta-Analyses

The Newcastle-Ottawa Scale (NOS)

We downloaded the following scale from: http://www.ohri.ca/programs/clinical_ epidemiology /oxford.asp, to evaluate the included studies qualities.

Newcastle-Ottawa Quality Assessment Scale: Cohort Studies

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Outcome categories. A maximum of two stars can be given for Comparability

Selection

1) Representativeness of the exposed cohort

a) truly representative of the average _____(describe) in the community ✱ □

b) somewhat representative of the average _____in the community ✱ □

c) selected group of users eg nurses, volunteers

d) no description of the derivation of the cohort

2) Selection of the non exposed cohort

a) drawn from the same community as the exposed cohort ✱

b) drawn from a different source

c) no description of the derivation of the non exposed cohort

3) Ascertainment of exposure

a) secure record (eg surgical records) ✱ □

b) structured interview □

c) written self report

d) no description

4) Demonstration that outcome of interest was not present at start of study

a) yes ✱

b) no

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Comparability

1) Comparability of cohorts on the basis of the design or analysis

a) study controls for _____ (select the most important factor) ✱

b) study controls for any additional factor ✱ (This criteria could be modified to indicate specific control for a second important factor.)

Outcome

1) Assessment of outcome

a) independent blind assessment ✱

b) record linkage □

c) self report

d) no description

2) Was follow-up long enough for outcomes to occur

a) yes (select an adequate follow up period for outcome of interest) ✱

b) no

3) Adequacy of follow up of cohorts

a) complete follow up - all subjects accounted for ✱

b) subjects lost to follow up unlikely to introduce bias - small number lost - > ___ % (select an adequate %) follow up, or description provided of those lost)✱

c) follow up rate < ____% (select an adequate %) and no description of those lost

d) no statement

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The Detailed Assessment Process of Every Included Article

Supplementary Table S1. Quality assessment of the studies based on Newcastle Ottawa scale.

Study(Year)	Selection 1)	Selection 2)	Selection 3)	Selection 4)	Comparability 1)	Outcome 1)	Outcome 2)	Outcome 3)	Score
Khasab (2016)	a	b	a	a	a	b	a	a	8
Perez-Miranda (2017)	a	a	a	a	a	b	b	a	7
Widmer (2019)	b	a	a	a	a	b	a	a	8
Marya (2020)	a	a	a	a	a	b	a	a	8
Bondi (2020)	b	a	a	a	a	b	b	a	7
Kouanda (2021)	b	a	a	a	a	b	a	a	8

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title page
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	2
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	2,3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3,4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	3
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	3
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5,6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency	5,6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	n/a
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7,9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7 and Figs
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	n/a
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	8
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	9
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	10
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	11

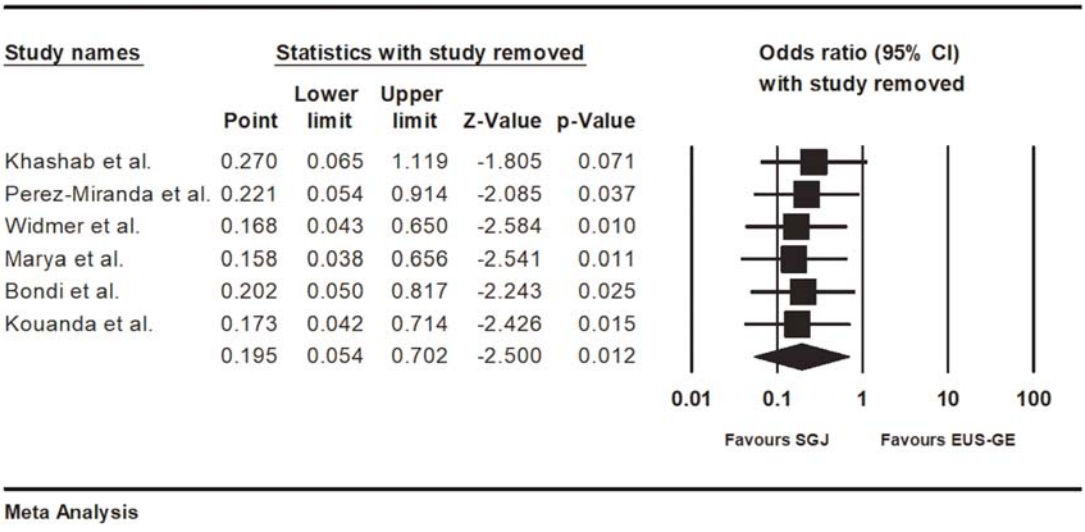
From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: www.prisma-statement.org.

Bomman Shivanand et al. Endoscopic ultrasound-guided gastroenterostomy… Endosc Int Open 2022; 10: E1–E12 | © 2022 The Author(s).

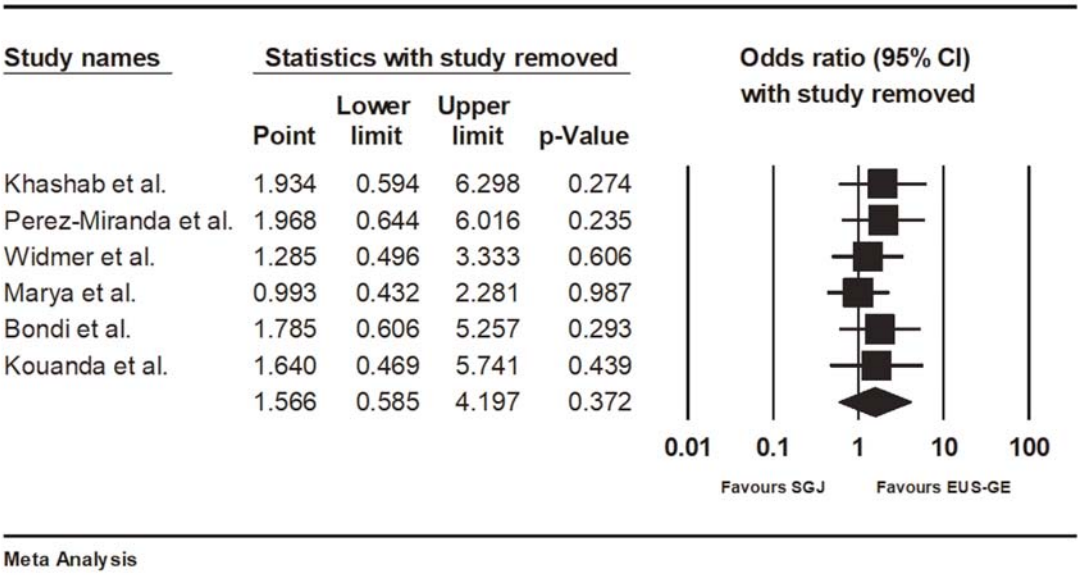
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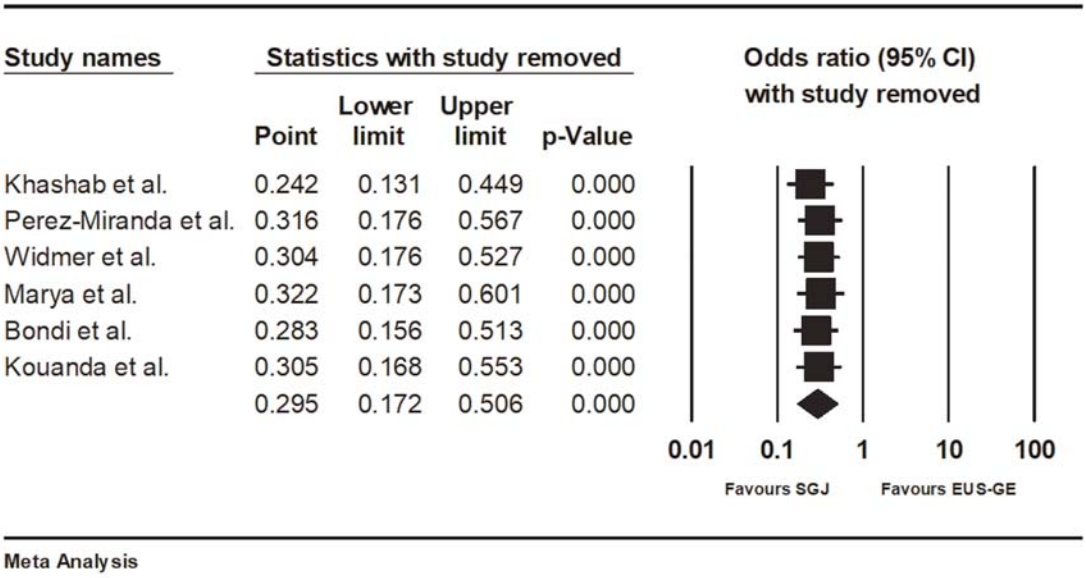
Supplemental Fig. 1 Sensitivity analysis: EUS-GE VS SGJ technical Success after "leave one study out" analysis. (Outcomes corresponding to a study represent the outcomes leaving that particular study out.)

Supplementary material



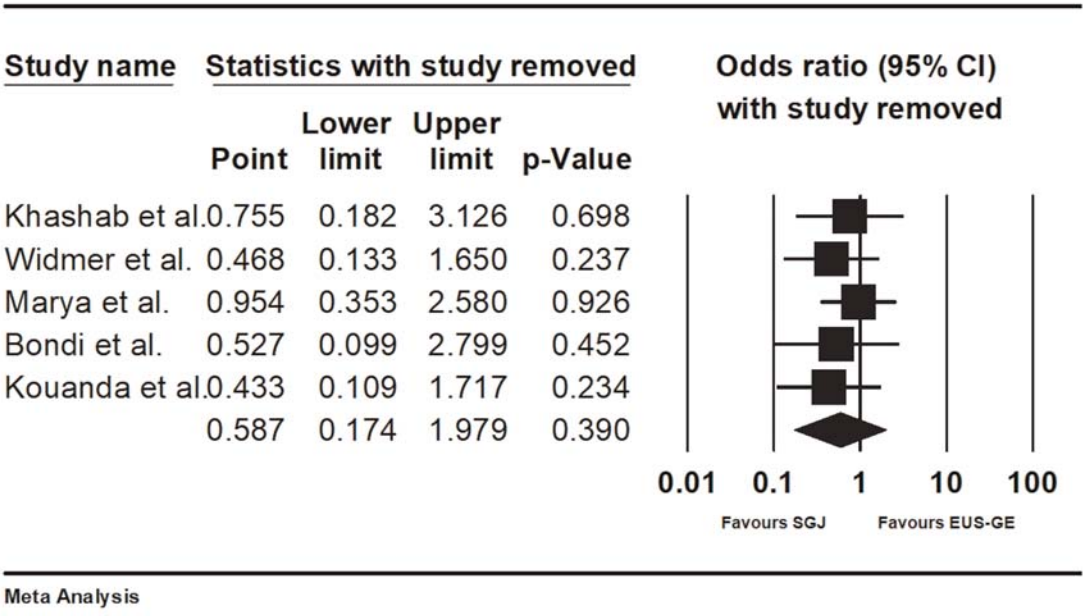
Supplemental Fig. 2 Sensitivity analysis: EUS-GE VS SGJ clinical success after "leave one study out" analysis. "Leave one study out" analysis. (Outcomes corresponding to a study represent the outcomes leaving that particular study out.)

Supplementary material



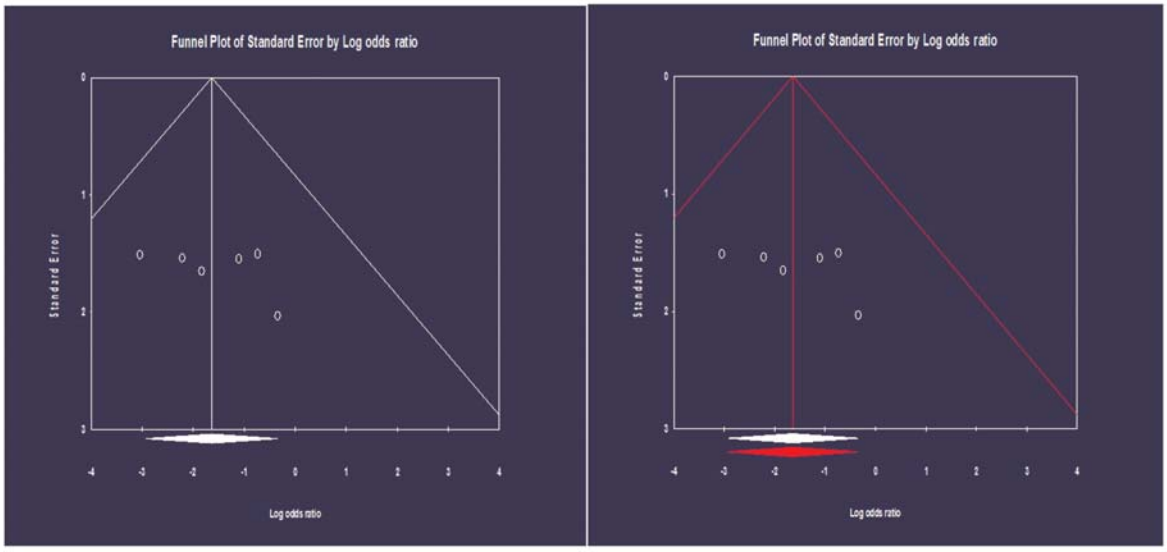
Supplemental Fig. 3 Sensitivity analysis: EUS-GE VS SGJ adverse events after "leave one study out" analysis. (Outcomes corresponding to a study represent the outcomes leaving that particular study out.)

Supplementary material



Supplemental Fig. 4 Sensitivity analysis: EUS-GE VS SGJ recurrence/reintervention rates after "leave one study out" analysis. (Outcomes corresponding to a study represent the outcomes leaving that particular study out.)

Supplementary material



Duval and Tweedie's trim and fill

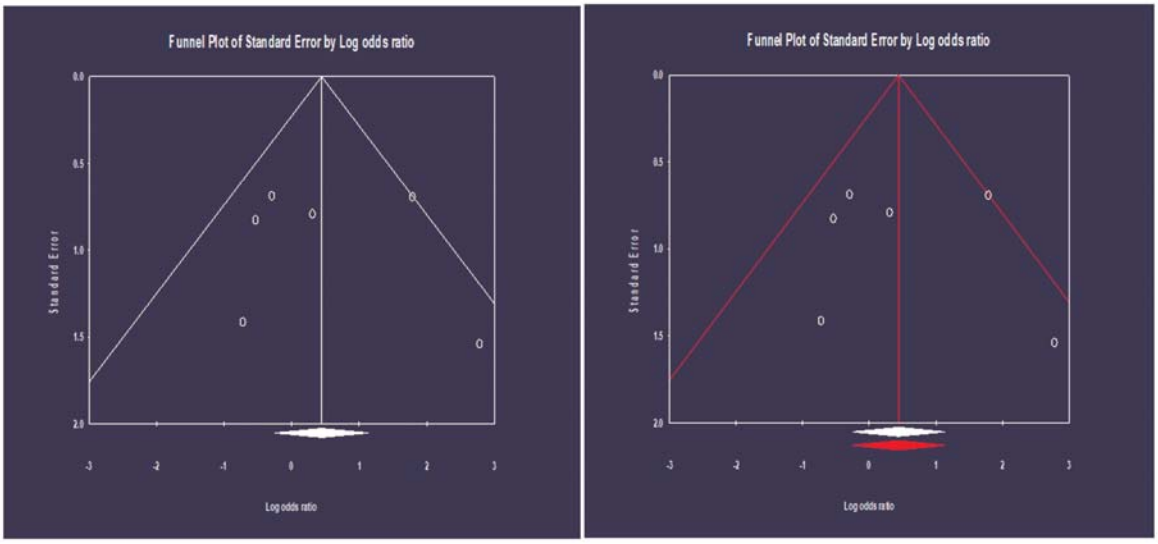
		Fixed Effects			Random Effects			Q Value
	Studies Trimmed	Point Estimate	Lower Limit	Upper Limit	Point Estimate	Lower Limit	Upper Limit	
Observed values		0.19464	0.05396	0.70208	0.19464	0.05396	0.70208	1.90942
Adjusted values	0	0.19464	0.05396	0.70208	0.19464	0.05396	0.70208	1.90942

Egger's regression intercept

Intercept	2.66361
Standard error	2.57999
95% lower limit (2-tailed)	-4.49958
95% upper limit (2-tailed)	9.82680
t-value	1.03241
df	4.00000
P-value (1-tailed)	0.18010
P-value (2-tailed)	0.36021

Supplemental Fig. 5 Funnel plot with trim and fill: technical success.

Supplementary material



Duval and Tweedie's trim and fill

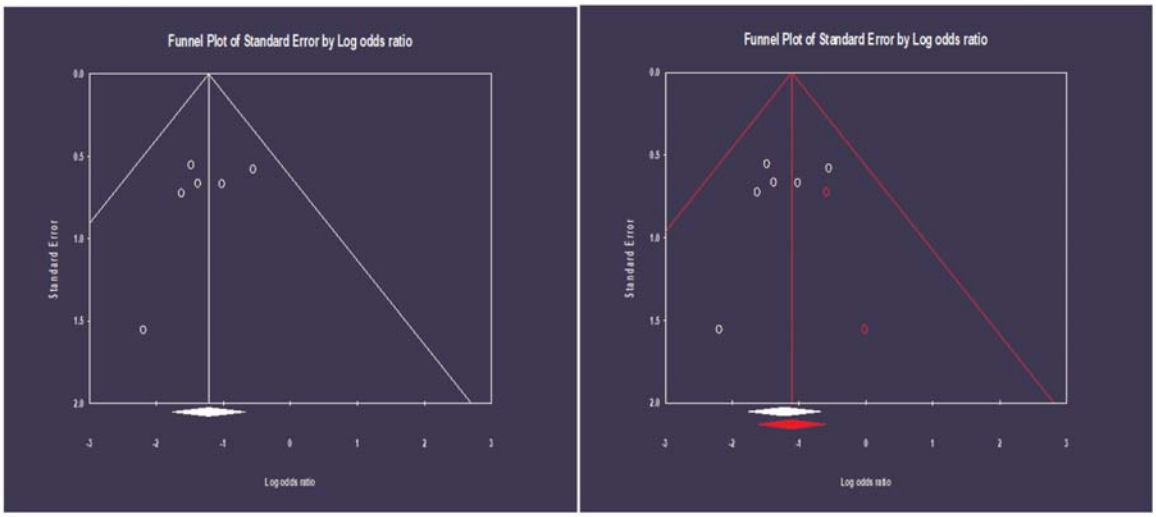
		Fixed Effects			Random Effects			Q Value
	Studies Trimmed	Point Estimate	Lower Limit	Upper Limit	Point Estimate	Lower Limit	Upper Limit	
Observed values		1.55424	0.78421	3.08040	1.56619	0.58449	4.19671	9.37838
Adjusted values	0	1.55424	0.78421	3.08040	1.56619	0.58449	4.19671	9.37838

Egger's regression intercept

Intercept	-1.20934
Standard error	1.07312
95% lower limit (2-tailed)	-4.18880
95% upper limit (2-tailed)	1.77012
t-value	1.12693
df	4.00000
P-value (1-tailed)	0.16140
P-value (2-tailed)	0.32280

Supplemental Fig. 6 Funnel plot with trim and fill: clinical success

Supplementary material



Duval and Tweedie's trim and fill

		Fixed Effects			Random Effects			Q Value
	Studies Trimmed	Point Estimate	Lower Limit	Upper Limit	Point Estimate	Lower Limit	Upper Limit	
Observed values		0.29495	0.17182	0.50634	0.29495	0.17182	0.50634	2.40874
Adjusted values	2	0.32914	0.20003	0.54159	0.32914	0.20003	0.54159	3.57000

Egger's regression intercept

Intercept	-1.20934
Standard error	1.07312
95% lower limit (2-tailed)	-4.18880
95% upper limit (2-tailed)	1.77012
t-value	1.12693
df	4.00000
P-value (1-tailed)	0.16140
P-value (2-tailed)	0.32280

Supplemental Fig. 7 Funnel plot with trim and fill: adverse events.